THE MALAWI NATIONAL EXAMINATIONS BOARD

2004 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

MATHEMATICS

Wednesday, 13 October **Subject Number: M131/I**

Time Allowed: 2 hours
8:30 – 10:30 am

PAPER I
(100 marks)

2004

Instructions

1. **This paper contains 6 pages. Please check.**
2. Answer **all** the **24** questions in this paper.
3. The maximum number of marks for each answer is indicated against each question.
4. Mathematical tables and answer books are provided.
5. Calculators may be used.
6. Used supplementary sheets must be handed in together with the answer book.
7. **All working must be clearly shown;** it should be done on the same sheet as the rest of the answer.
8. Write your **Examination Number** at the top of each page of your Answer Book.

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Answer **all** the **twenty four** questions in this **paper.**

1. Factorise completely $6+x-2x^2$. $(-2x(-3)(x-2)$ **(4 marks)**

2. A straight line passes through the point (1, 6) and cuts the y-axis at 4, calculate its gradient. **(4 marks)**

3. Given that $g(x)=\frac{3x}{x+1}$, calculate the value of x when $g(x)=2$. $3x=2x+2$ **(4 marks)**

4. A quantity b varies jointly with r and t, and $b = 108$ when $r = 3$ and $t = 6$. Find an equation which expresses b in terms of r and t. $3a+6b=108$ **(4 marks)**

5. Given that $C' = (5, -5)$ is the image of C (1, 1) after translation, find the translation vector that maps C into C'. **(4 marks)**

6. In **Figure 1**, **ABC** is a triangle in which angle **BAC** = 90°, angle **ABC** = 27° and **AB** = 5 cm.

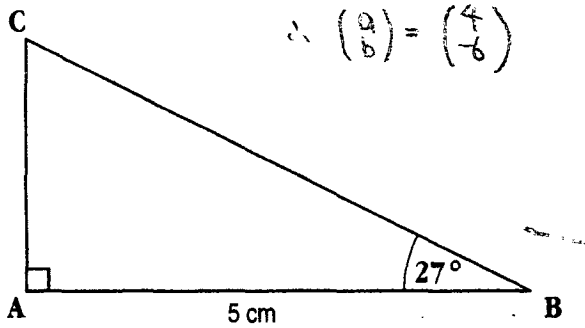

Figure 1

Calculate the length of **AC**. **(4 marks)**

7. Make y the subject of the formula

$$\frac{y+d}{c}=\frac{3d}{y-d}$$

(5 marks)

8. Given that $\begin{pmatrix}3c & c\\ 5 & 1\end{pmatrix}\begin{pmatrix}4\\2\end{pmatrix}=\begin{pmatrix}28\\22\end{pmatrix}$, find c. **(4 marks)**

9. The sum of the first two terms of a geometric progression is 4. If the first term is 3, find the common ratio. **(4 marks)**

10. The areas of two similar triangles **ABC** and **HKL** are 100 cm^2 and 256 cm^2 respectively. If the length of **AB** is 5 cm, calculate the length of **HK**. **(5 marks)**

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11. Figure 2 shows an unshaded region bounded by three inequalities.

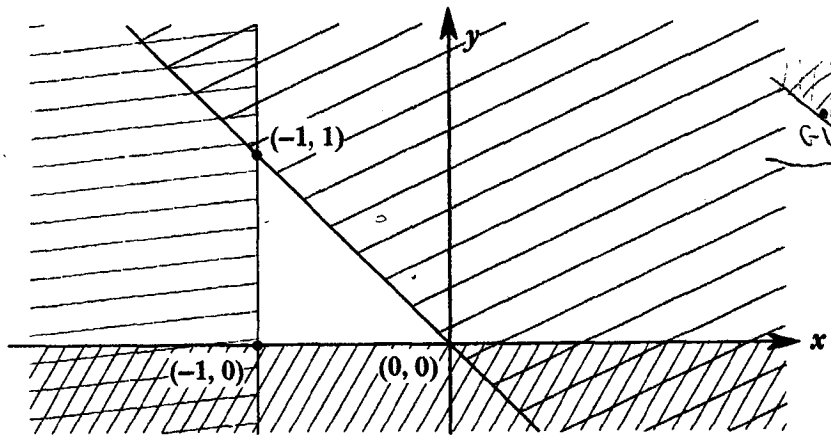

Figure 2

Write down the three inequalities. **(5 marks)**

$y = ax + b$

$1 = -a + b$

$0 = b$

$1 = -a$

$a = -1$

$a = -1$

$b = 0$

$y = -x$

$y > -x$

$y < -x$

$1 > -1$

$y > -x$

12. **Figure 3** shows the speed time graph of a moving object.

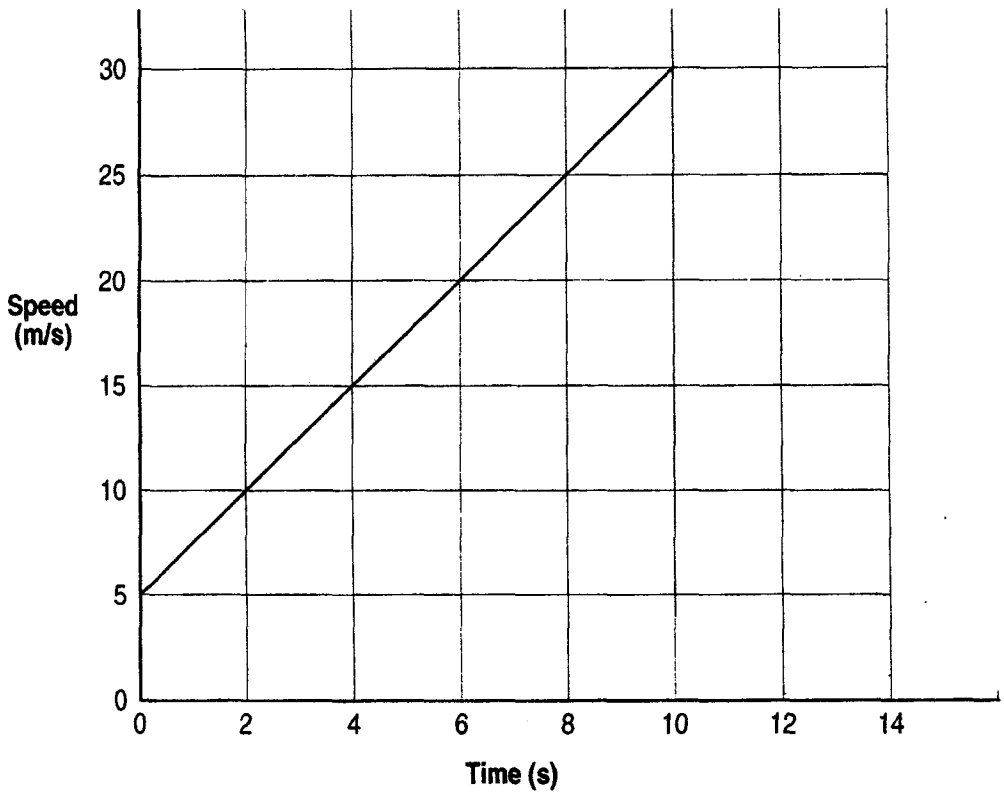

Figure 3

$\frac{10-5}{2-0} = 2.5 \text{ m/s}^2$

Use the graph to find the acceleration of the object. **(3 marks)**

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13. A chord is 6 cm away from the centre of a circle. If the chord is 16 cm long, calculate the radius of the circle. **(5 marks)**

14. Simplify $\dfrac{\sqrt{2}}{\sqrt{2}-1}$. $= \dfrac{\sqrt{2}(\sqrt{2}+1)}{2-1} = 2+\sqrt{2}$ **(4 marks)**.

15. Given that $\overrightarrow{AB} = \begin{pmatrix}3\\5\end{pmatrix}$, and $\overrightarrow{BC} = \begin{pmatrix}9\\15\end{pmatrix}$, show that points **A, B** and **C** are collinear. **(3 marks)**

$\overrightarrow{BC} = 3\left(\frac{3}{5}\right) = 3\overrightarrow{AB}$

16. Express $\dfrac{m-2}{m-3} + \dfrac{m+3}{m+2}$ as a single fraction in its lowest term. **(5 marks)**

$\dfrac{m^2-4+m^2-9}{(m-3)(m+2)} = \dfrac{2m^2+5}{m^2-m-6}$

17. In **Figure 4**, **TX** and **TY** are tangents to the circle **XHY** at **X** and **Y**.

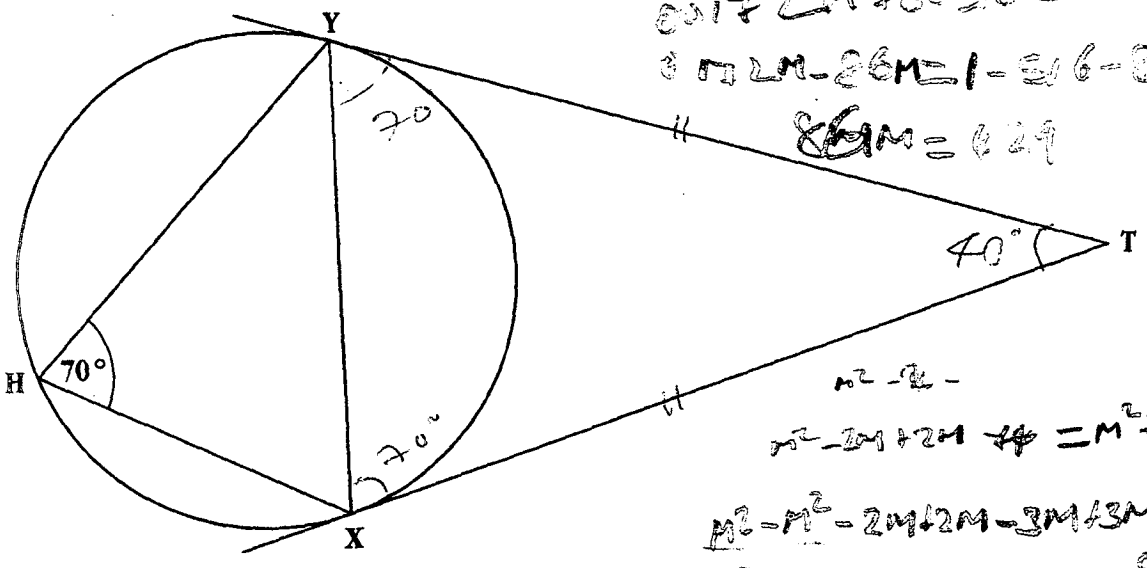

Figure 4

If angle **XHY** = 70°, calculate angle **XTY**. **(4 marks)**

18. Use the remainder theorem to prove that $(x-y)$ is a factor of a polynomial $x^2(y-2)+y^2(2-x)$. **(5 marks)**

$= x^2y-2x^2+2y^2-xy^2$
$= xy(x-y)-2(x+y)(x-y)$

19. Solve the equation $\log_{10}(2m+6) = 1+\log_{10}(m-1)$. **(5 marks)**

$2m+6 = 10m-10$
$-8m = -16$
$m = 2$

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20. In **Figure 5**, the histogram is representing the ages of the people who attended a meeting.

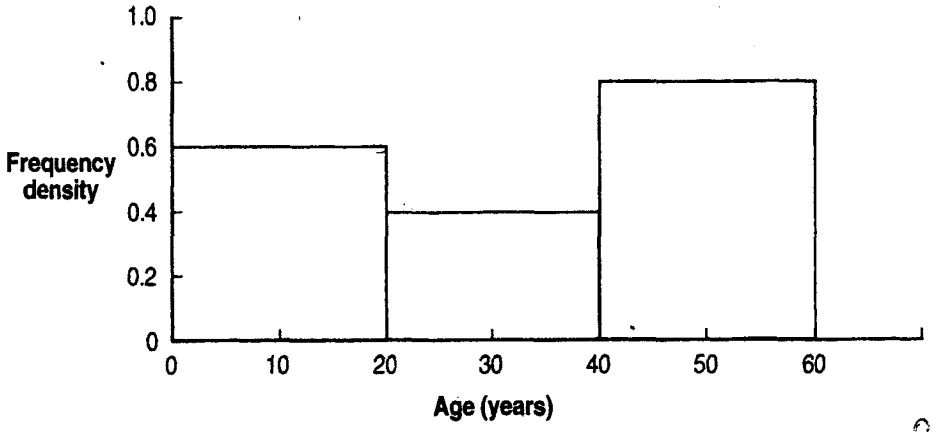

Figure 5

0.6 × 3 = 1.3
0.4
2.4

0.6
0.6
0.4
0.4
0.8
0.8

Calculate the number of people who were between 20 and 40 years old at the meeting. **(3 marks)**

21. **Figure 6** shows a triangular prism, in which **BC** = 3 cm, **EF** = 4 cm, **BF** = 6 cm and angle **ABC** = 90°.

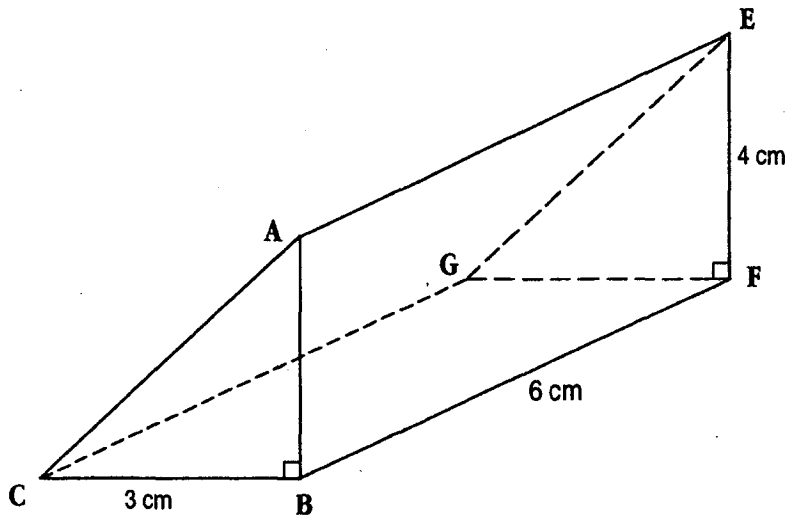

Figure 6

Calculate the volume of the prism. **(3 marks)**

$\frac{1}{2} \times 3 \times 4 \times 6 = 36$ cm³ //

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22. Given that the Universal set $\xi = \{11, 14, 15, 17, 18, 20, 23, 26\}$, set $X = \{11, 14, 15, 17, 18, 20\}$, and set $Y = \{15, 17, 18, 20, 23, 26\}$, find $X' \cup Y'$. **(4 marks)**

23. In a plastic bag there are x blue pens, 6 black pens and 4 red pens. If the probability of picking a red pen is $\frac{1}{5}$; calculate the number of blue pens. **(4 marks)**

$\frac{4}{x+6+4} = \frac{1}{5}$

$\frac{4}{10+x} = \frac{1}{5}$

$\therefore 10+x = 20$

$\therefore x = 10$

24. Solve the simultaneous equations

$y = x^2$

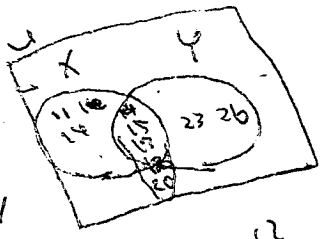
$y = 5x - 6.$ **(5 marks)**

$x^2 - 5x + 6 = 0$

$(x-2)(x-3) = 0$

$x = 2,\ y = 4$

$x = 3,\ y = 9$



$X' = \{23, 26\}$

$Y' = \{11, 14\}$

$X' \cup Y' = \{23, 26, 11, 14\}$

END OF QUESTION PAPER

NB: This paper contains 6 pages.